Polar Coordinates & Polar Graphs

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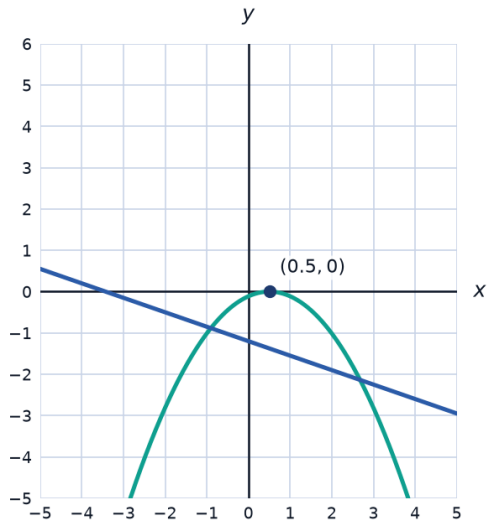
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Analyze the polar curve $r(\theta) = 2 - 4\cos\theta$ for $0 \le \theta < 2\pi$. Identify the curve type, find the maximum distance from the pole, and determine the exact angles θ where the curve passes through the pole.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the exact angle θ between $\mathbf{u} = \langle 3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2 \rangle$, and compute the vector projection $\text{proj}_{\mathbf{v}}\mathbf{u}$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A cup of coffee at $92\,^\circ\text{C}$ is placed in a room kept at a constant $20\,^\circ\text{C}$. After 10 minutes, the coffee has cooled to $68\,^\circ\text{C}$. Using Newton's Law of Cooling $T(t) = 20 + 72b^{t/10}$, find the exact temperature after 20 minutes and when the coffee reaches $36\,^\circ\text{C}$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given that $\cos\alpha = -\frac{5}{13}$ with $\frac{\pi}{2} < \alpha < \pi$, find the exact values of $\sin(2\alpha)$, $\cos(2\alpha)$, and $\tan(2\alpha)$.

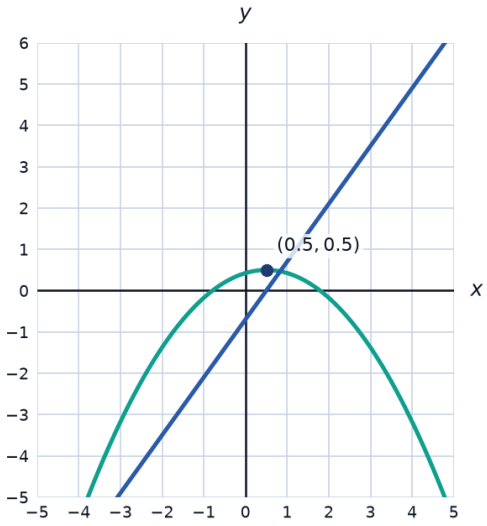
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the exact value of $\sin\left(\arccos\left(-\frac{3}{5}\right)\right)$ and $\tan\left(\arcsin\left(\frac{8}{17}\right)\right)$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For a quadratic function $f(x)$, selected values are $f(1) = 4$, $f(2) = 9$, $f(3) = 18$, and $f(4) = 31$. Compute the first and second differences, verify that f is quadratic, and find its explicit formula $f(x) = ax^2 + bx + c$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $f(x) = \frac{4e^{2x}}{e^{2x} + 3}$. Prove that f is strictly one-to-one and find an explicit formula for its inverse function $f^{-1}(x)$, stating the domain of f^{-1}.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Identify the conic section $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ by completing the square. Find its center, vertices, foci, and asymptotes.

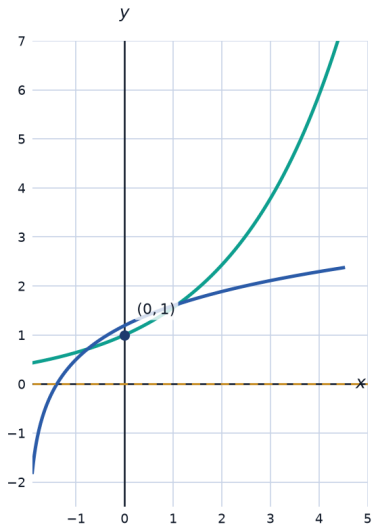
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the logarithmic equation $\log_4(x + 6) + \log_4(x - 6) = 3$, checking for extraneous roots.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Evaluate $\log_2(5) \cdot \log_5(9) \cdot \log_3(16)$ without a calculator using the change-of-base formula.

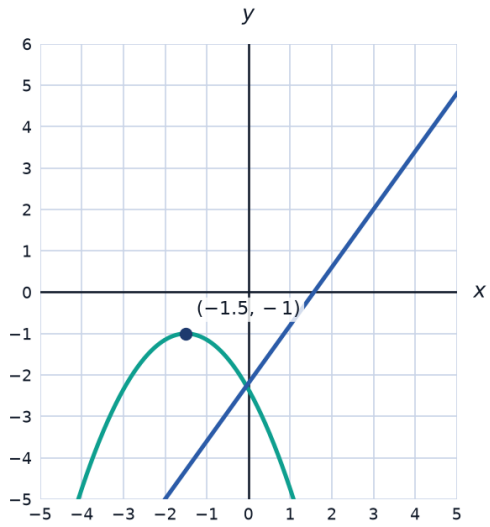


11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Let $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x+5}$. Find the composite function $(f \circ g)(x)$ and state its exact domain.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).



13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

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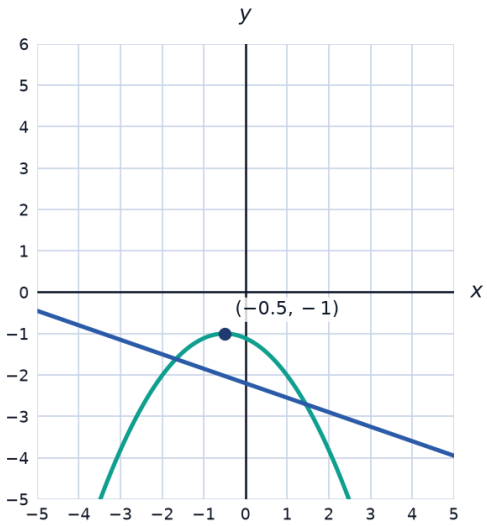
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.03 Rates of Change & Concavity**

Convert the polar equation $r = 8\sin\theta$ into rectangular coordinates, and find its center and radius.

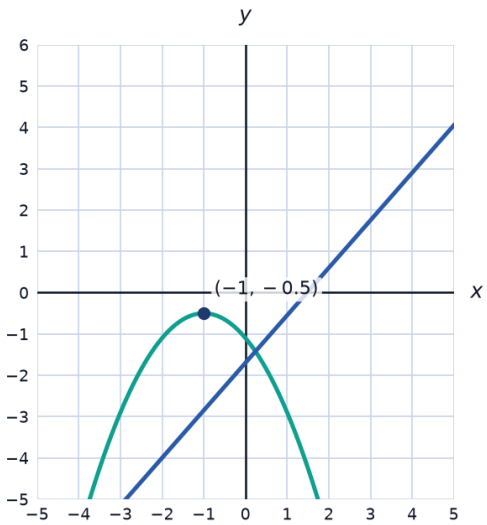
16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 2.01 Exponential Functions & Growth Models**

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.02 Logarithmic Functions & Properties**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.



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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Inverse & Composite Functions

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x - 1}{2x + 5}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar point $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

a. $(-3\sqrt{3}, 3)$

b. $(3\sqrt{3}, -3)$

c. $(-3, 3\sqrt{3})$

d. $(3, -3\sqrt{3})$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular point $(-4, 4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $\left(4\sqrt{2}, \frac{3\pi}{4}\right)$

b. $\left(4\sqrt{2}, \frac{7\pi}{4}\right)$

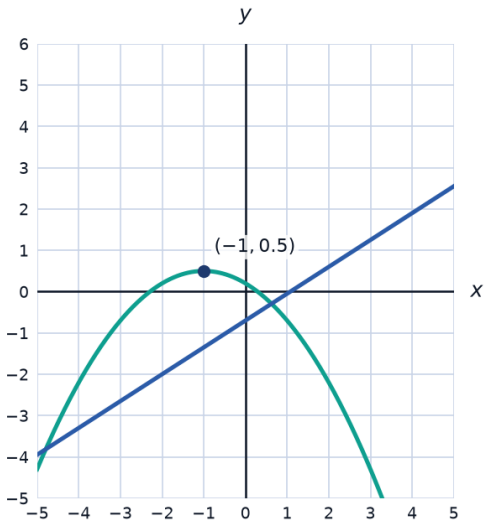
c. $\left(8, \frac{3\pi}{4}\right)$

d. $\left(4\sqrt{2}, \frac{5\pi}{4}\right)$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which polar point represents the same location as $(r, \theta) = (3, \frac{\pi}{3})$?



a. $(-3, \frac{4\pi}{3})$

b. $(-3, \frac{\pi}{3})$

c. $(3, -\frac{\pi}{3})$

d. $(-3, \frac{2\pi}{3})$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar equation $r = 6\cos\theta$ into rectangular coordinates.

a. $(x-3)^2 + y^2 = 9$

b. $x^2 + (y-3)^2 = 9$

c. $x^2 + y^2 = 6x$

d. $x^2 + y^2 = 36$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular equation $x^2 + y^2 = 16$ to polar form.

a. $r = 4$

b. $r = 16$

c. $r = 4\cos\theta$

d. $r = 16\sin\theta$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $r = 4\csc\theta$ to rectangular coordinates.

a. $y = 4$

b. $x = 4$

c. $y = 4x$

d. $x^2 + y^2 = 16$

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 5\cos(4\theta)$ have?

a. 8

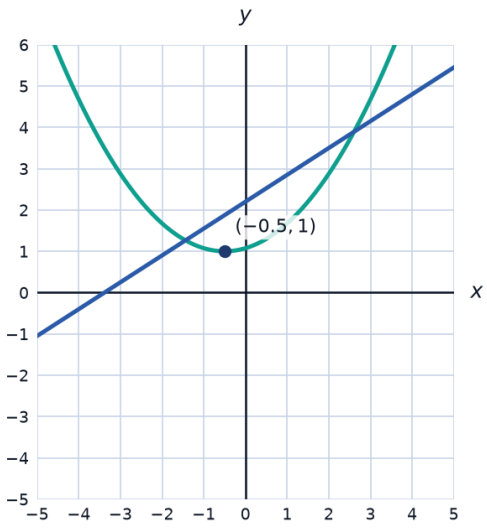
b. 4

c. 16

d. 2

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the maximum distance from the pole (origin) to the curve $r = 4\sin(3\theta)$?



a. 4

b. 12

c. 3

d. 8

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Classify the polar curve $r = 2 + 2\cos\theta$.

a. Cardioid (heart-shaped limaçon)

b. Limaçon with an inner loop

c. Dimpled limaçon

d. Convex limaçon

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which polar equation produces a limaçon with an inner loop?

a. $r = 2 + 4\cos\theta$

b. $r = 4 + 2\cos\theta$

c. $r = 3 + 3\sin\theta$

d. $r = 5 + 4\cos\theta$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

At what angles $\theta \in [0, 2\pi)$ does the curve $r = 2 + 4\cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$

b. $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

c. $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

d. $\theta = \pi$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

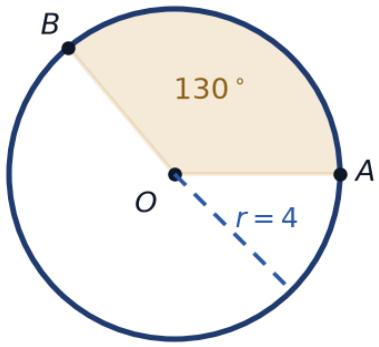
c. $2\sqrt{3}$

d. $-2\sqrt{3}$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.



a. $\frac{\pi}{2}$

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. 4

c. $2\sqrt{2}$

d. 1

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. 2

b. 4

c. 8

d. 16

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. 5

b. 7

c. 25

d. 16

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Inverse & Composite Functions

What is the Cartesian equation of the polar curve $r = 1$?

a. $x^2 + y^2 = 1$

b. $x^2 - y^2 = 1$

c. $x^2 + y^2 = 2$

d. x+y=1

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

The polar equation $r = 2\cos\theta$ represents a circle. What is its radius?

a. 2

b. 3

c. 5

d. 1

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.03 Parametric Equations & Modeling

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. There exists $c \in (1, 4)$ with $f(c) = 0$

c. $f(2) = 1$

d. f is constant

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Find θ (in $[0^\circ, 360^\circ)$) for the Cartesian point $(3, 0)$.

a. 180

b. 0

c. 181

d. 90

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Convert the polar point $(4, 30^\circ)$ to Cartesian coordinates.

a. $(2\sqrt{3}, 3)$

b. $(-2\sqrt{3}, -2)$

c. $(1 + 2\sqrt{3}, 2)$

d. $(2\sqrt{3}, 2)$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Rates of Change & Concavity

How many petals does the rose $r = \cos(2\theta)$ have?

a. 4

b. 6

c. 2

d. 5

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Find the area of the polar region bounded by $r = 2$ for $0 \le \theta \le \frac{\pi}{6}$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{4\pi}{3}$

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